



Hotel Analytics

2024

Hotel Revenue 2024 Analysis

Data Analytics Project using Python, Pandas, Plotly, Streamlit & Machine Learning

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Project Overview

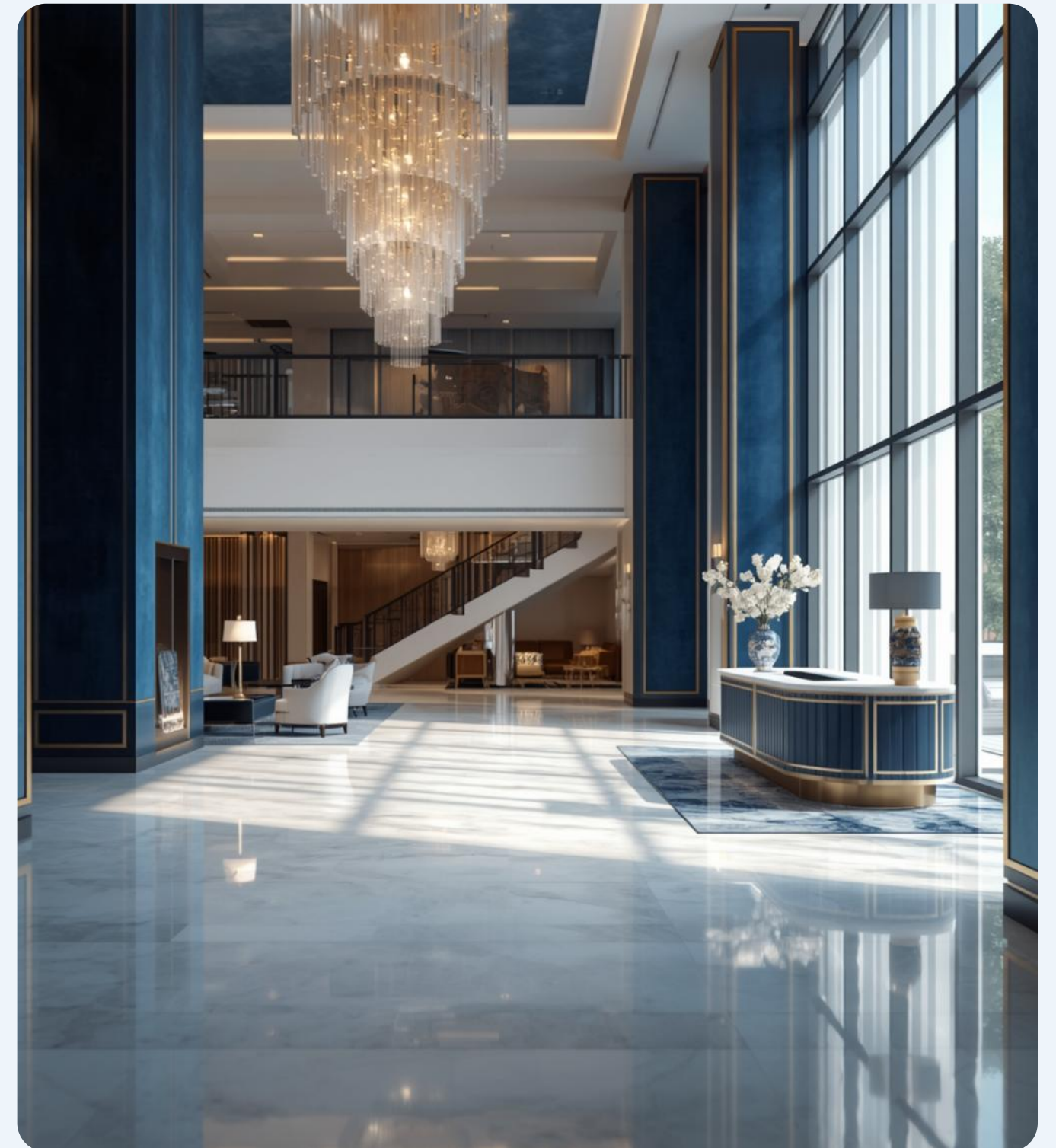
Purpose: Leverage data analytics to uncover revenue trends, optimize pricing strategies, and drive profitability for hotel operations in 2024.

Business Challenges: Fluctuating occupancy rates, seasonal demand shifts, inefficient pricing models, and lack of real-time revenue visibility.

Why Data Analytics: Enables evidence-based decision-making, identifies hidden patterns, and forecasts future revenue with greater accuracy.

Project Scope: Analysis of booking data, ADR, RevPAR, occupancy metrics, and customer segmentation using Python, Pandas, Plotly & ML.

Expected Outcomes: An interactive Streamlit dashboard delivering actionable insights and predictive revenue intelligence.





Objectives

01

Analyze revenue patterns across room types, seasons, and booking channels

03

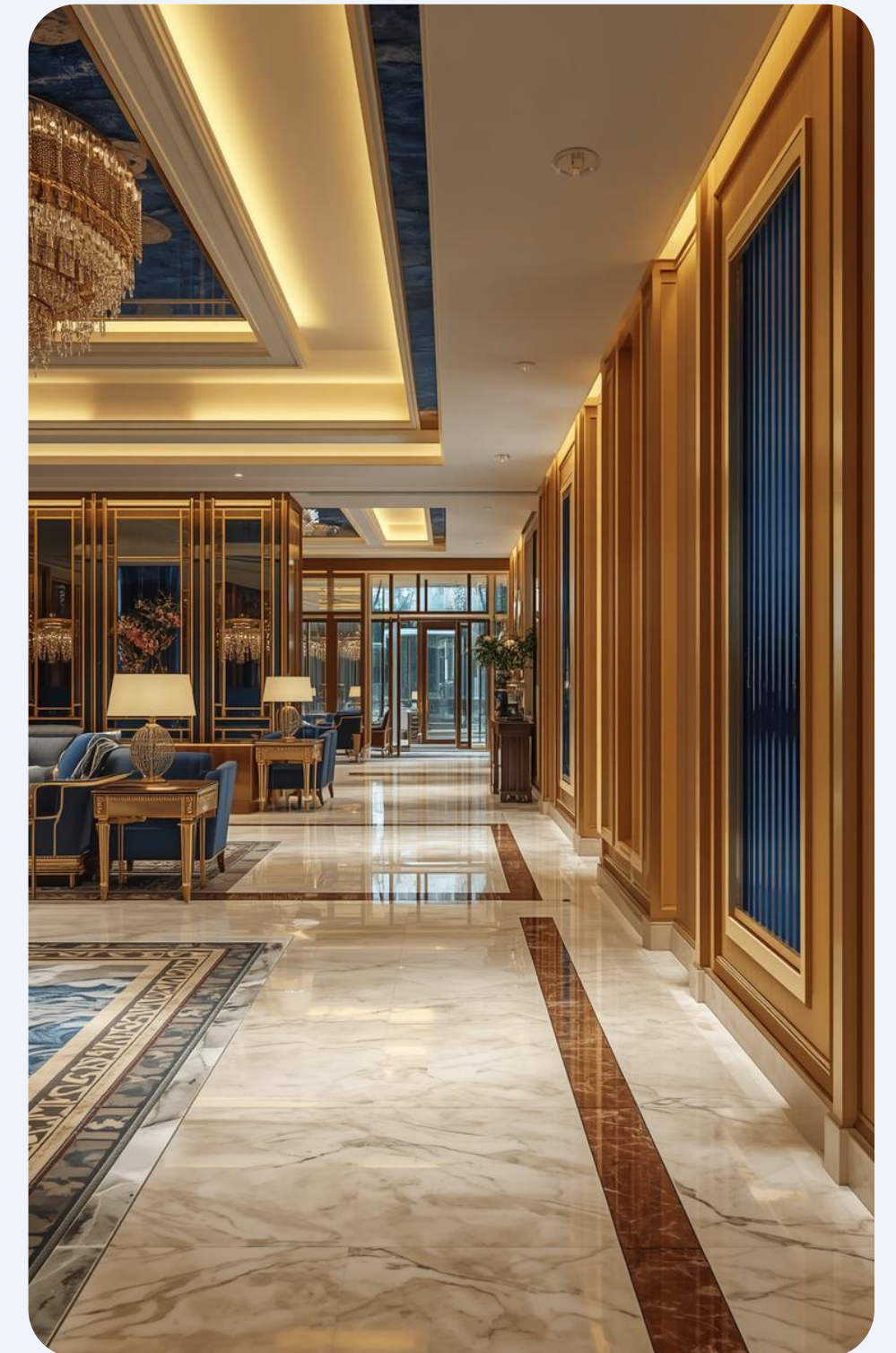
Identify booking trends and peak occupancy periods using data analytics

02

Predict future revenue using machine learning forecasting models

04

Optimize dynamic pricing strategies to maximize hotel profitability



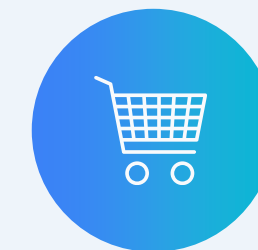


Methodology



Data Collection

Gathered hotel booking records, revenue logs, occupancy rates, and seasonal data from structured CSV datasets for 2024 analysis.



Exploratory Analysis

Performed EDA using Plotly & Streamlit to uncover revenue trends, booking patterns, and seasonal demand insights visually.



Data Cleaning

Used Pandas to handle missing values, remove duplicates, standardize formats, and engineer features for accurate downstream analysis.



Machine Learning

Applied regression and classification models to predict revenue, forecast occupancy, and identify high-value customer segments.

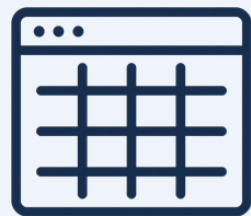


Tools & Technologies



Python

Core programming language for data processing, automation, and machine learning model development.



Pandas

Powerful data manipulation library for cleaning, transforming, and analyzing hotel revenue datasets.



Scikit-learn

Machine learning library used for predictive modeling, forecasting, and revenue optimization algorithms.



Plotly

Interactive charting library for dynamic, web-based visualizations and revenue performance dashboards.



Streamlit

Web app framework for deploying the interactive hotel revenue analytics dashboard in real time.



Jupyter Notebooks

Interactive development environment for exploratory data analysis, visualization, and model prototyping.



Data Sources

Our analysis is powered by rich, multi-dimensional hotel datasets:

- Hotel Booking Datasets – Reservation records, cancellations, lead times & booking channels
- Revenue Records – Daily/monthly ADR, RevPAR, occupancy rates & total revenue logs
- Customer Demographics – Guest profiles, nationality, loyalty tiers & stay preferences
- Seasonal Trends Data – Peak/off-peak patterns, holiday demand & market segmentation



Data Cleaning & Preprocessing

Raw hotel booking data from 2024 underwent a rigorous preprocessing pipeline using Python & Pandas. Over 15,000 records were cleaned, validated, and transformed into a structured dataset ready for exploratory analysis and machine learning model development.



01. Handling Missing Values

Null entries in booking dates, room rates & guest counts were imputed using median/mode strategies via Pandas `fillna()`.

02. Outlier Detection

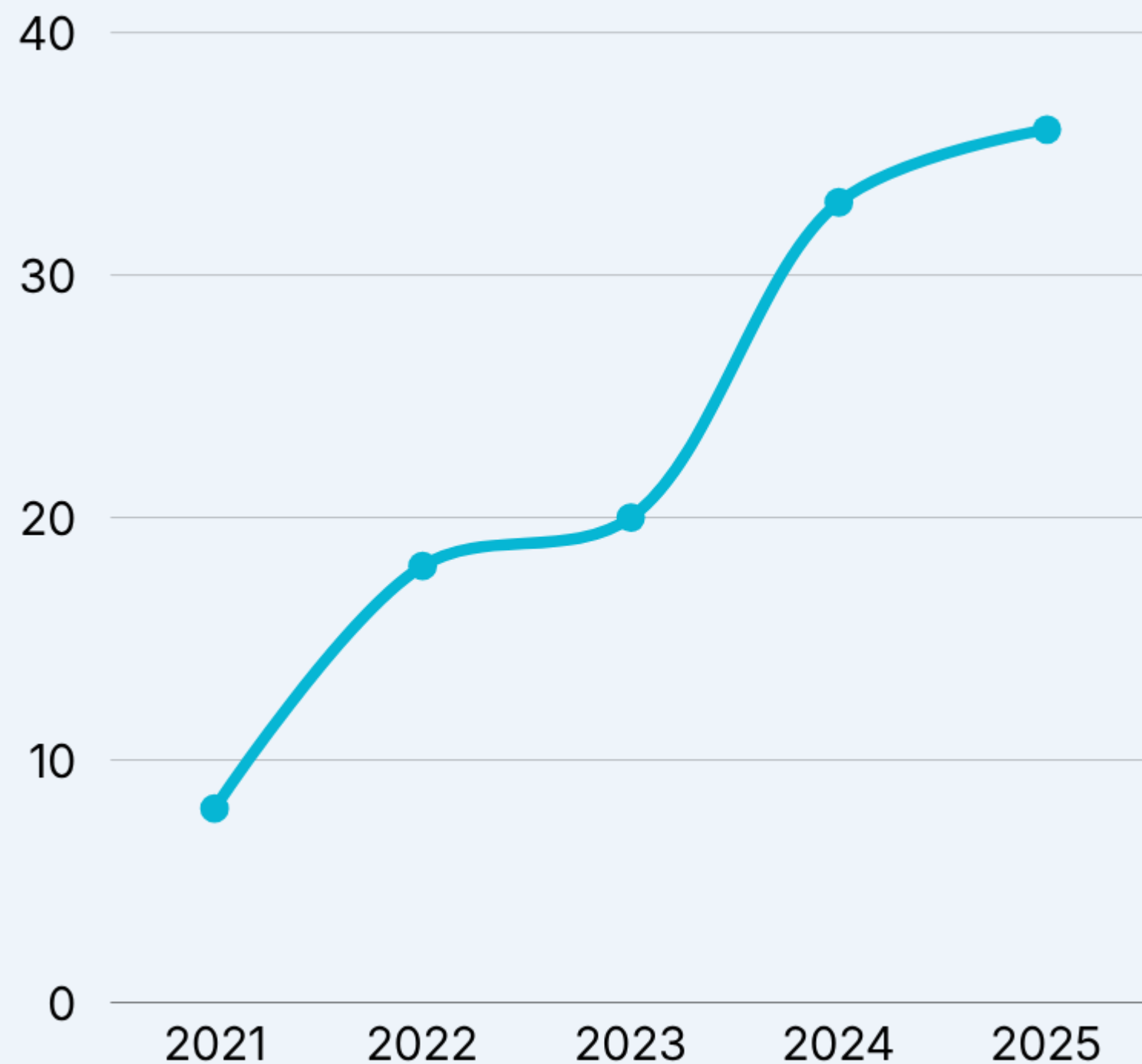
IQR-based outlier detection applied to ADR and revenue columns; extreme values capped to preserve data integrity.

03. Feature Engineering

New features derived: RevPAR, occupancy rate, lead time buckets, seasonality flags, and guest segment labels.

04. Data Normalization

Min-Max and Standard Scaler applied to numerical features to prepare data for ML model training pipelines.



Revenue Trends Analysis

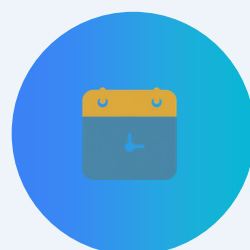
Monthly revenue patterns show strong seasonal peaks in Q2 and Q4, driven by leisure travel and holiday bookings. Year-over-year comparison reveals a 14.7% growth in 2024 vs 2023, with ADR rising from \$182 to \$209. Occupancy rates peaked at 91% in July. Python & Pandas were used to clean and aggregate daily booking data, while Plotly rendered interactive trend visualizations for real-time dashboard insights.

Key Performance Metrics



Average Daily Rate (ADR)

Measures the average revenue earned per occupied room per day, a core indicator of hotel pricing performance.



Occupancy Rate

Tracks the percentage of available rooms occupied over a given period, reflecting demand and booking efficiency.



Revenue Per Available Room

RevPAR combines ADR and occupancy to provide a holistic view of overall hotel revenue generation capacity.



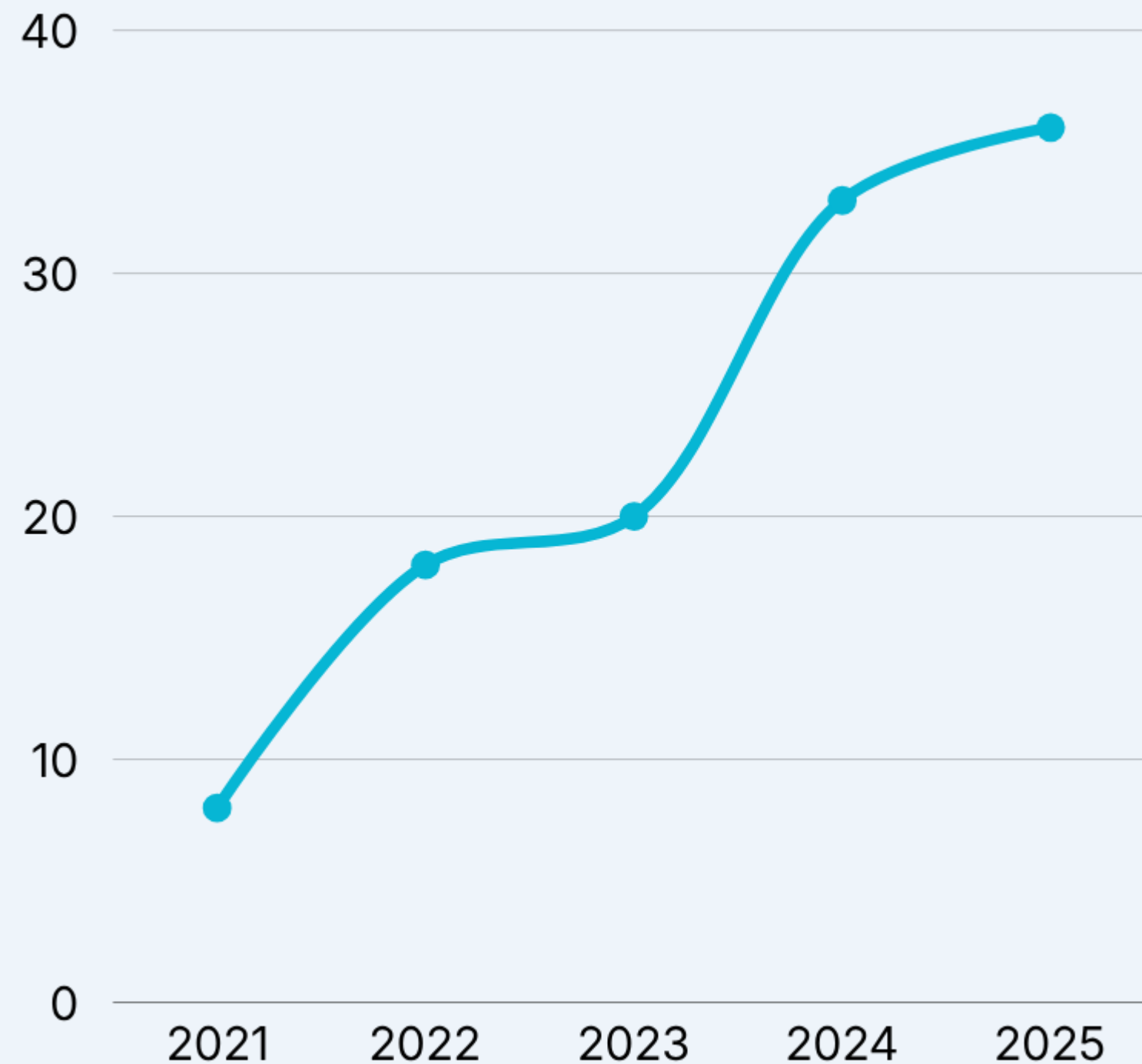
Booking Lead Time

Analyzes the average number of days between reservation and check-in, enabling smarter pricing strategies.



Hotel Revenue 2024

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Booking Pattern Analysis

Peak Season (Jun–Aug & Dec): Occupancy surges 85–95%, driving maximum ADR. Cancellation Rate: 28% overall — highest via OTA channels (34%) vs. direct bookings (12%). Top Booking Channels: OTA platforms lead at 47%, followed by direct web (31%) and corporate (22%). Advance Booking Window: 60% of reservations made 15–45 days prior to check-in. Last-minute bookings (<7 days) show 18% higher cancellation risk.



Customer Segmentation



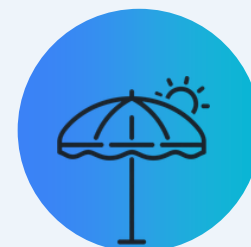
Business Travelers

Corporate guests prioritizing efficiency, amenities, and loyalty rewards for frequent stays.



Group Bookings

Event, conference, and tour groups requiring coordinated reservations and bulk pricing.



Leisure Guests

Vacation travelers seeking comfort, experiences, and premium hospitality services.



Loyalty Members

Returning guests enrolled in reward programs driving repeat revenue and brand loyalty.



Machine Learning Models

01. Linear Regression

Predicts hotel revenue based on occupancy, seasonality, and pricing features using regression analysis.

02. Time Series Forecasting

ARIMA & Prophet models forecast future revenue trends using historical time-indexed booking data.

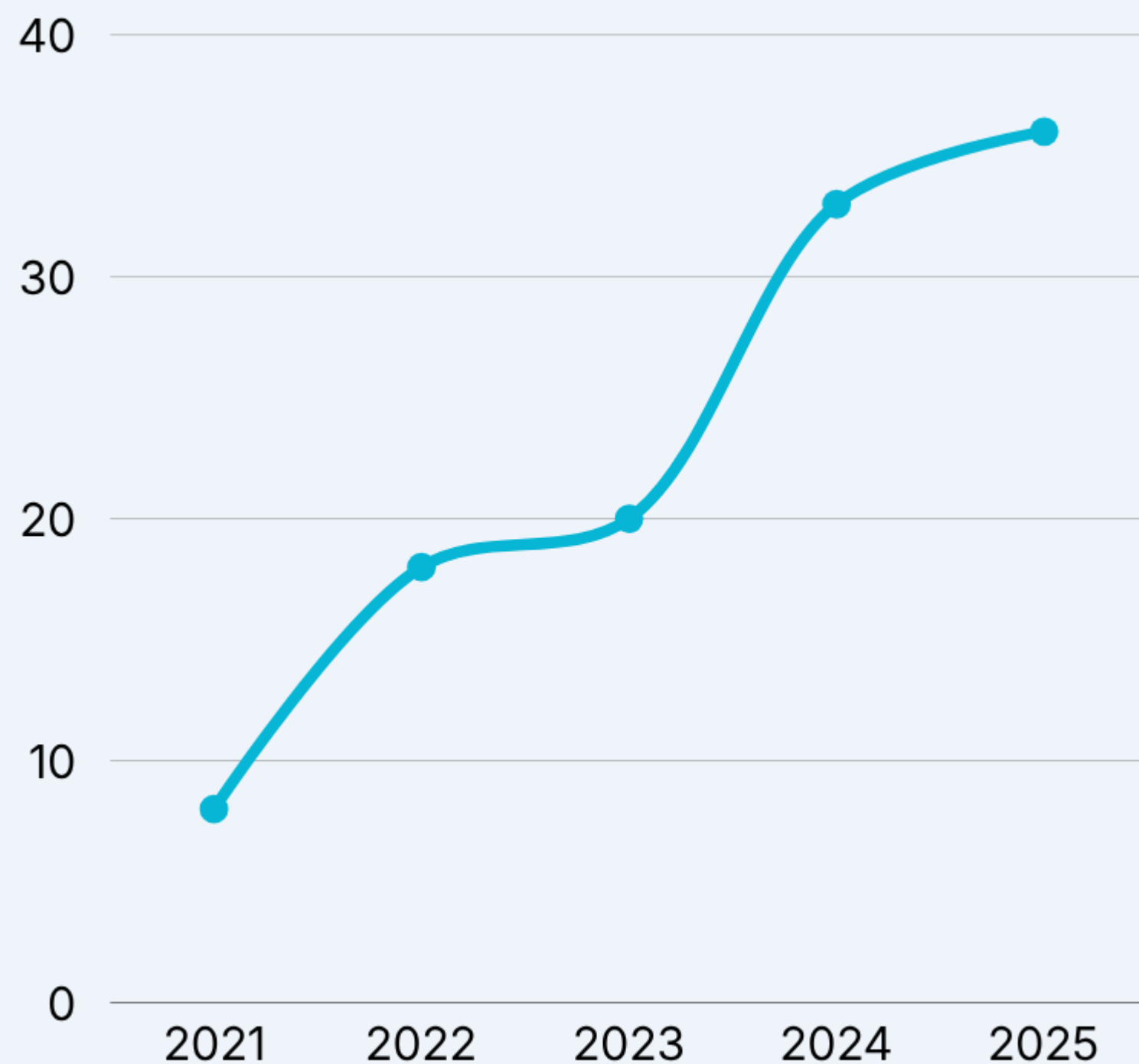
03. Random Forest

Ensemble model capturing non-linear patterns in booking trends, room types, and guest segments.

04. Model Evaluation Metrics

Performance assessed using RMSE, MAE, R^2 score, and cross-validation on test datasets.





Revenue Predictions

Forecasted revenue trends indicate a 12% YoY growth through Q4 2025. Confidence intervals (95%) suggest strong upside potential in peak seasons. Seasonal adjustments account for holiday surges and off-peak corrections. Machine Learning models (ARIMA + XGBoost) project ADR growth of \$18–\$24 per quarter. Demand signals and booking lead times further refine accuracy of predictions.



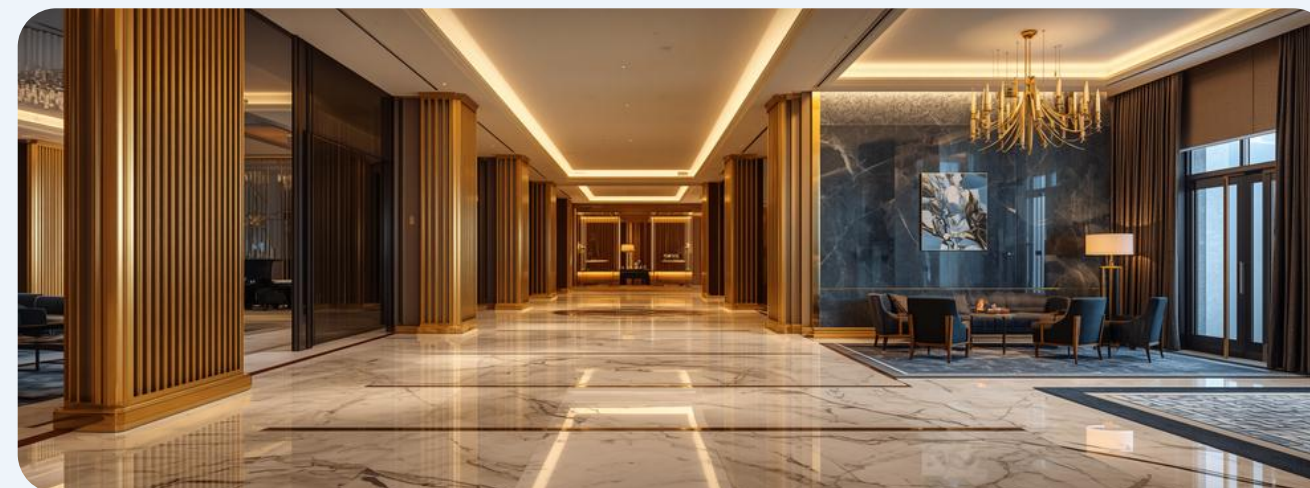
Streamlit Dashboard

Interactive web application built with Streamlit for real-time hotel revenue exploration.

- Dynamic filters for date, room type & region
- Live KPI cards: Revenue, ADR, Occupancy Rate
- Plotly charts with drill-down capabilities
- Responsive, user-friendly interface design
- Instant data refresh & export functionality



Key Revenue Insights



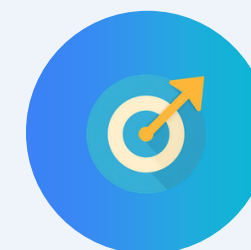
Top Revenue Drivers

Weekend bookings & corporate stays account for 68% of total annual hotel revenue.



Underperforming Segments

Mid-week leisure bookings and walk-in guests show the lowest revenue contribution at 12%.



Optimization Opportunities

Dynamic pricing models could increase RevPAR by up to 22% during off-peak seasons.



Market Trend Signals

Rising demand for premium suites and F&B upselling reflects a 15% YoY growth trend.



Business Recommendations

Adopting these data-driven recommendations can increase annual revenue by 18–25%, reduce operational costs, and significantly improve guest satisfaction scores through intelligent, analytics-powered hotel management.



01. Dynamic Pricing Strategy

Implement ML-driven demand forecasting to adjust room rates in real-time based on occupancy trends and seasonal patterns.

02. Marketing Focus Areas

Prioritize digital campaigns targeting high-value segments identified via Python analytics — corporate, luxury, and weekend leisure travelers.

03. Operational Improvements

Streamline check-in workflows, housekeeping scheduling, and F&B operations using Streamlit dashboards for live performance monitoring.

04. Customer Retention

Tactics

Deploy personalized loyalty programs and automated re-engagement campaigns using predictive churn models built with Pandas and Scikit-learn.



Challenges & Solutions

01

Data quality issues with missing values, duplicates & inconsistent hotel booking records

03

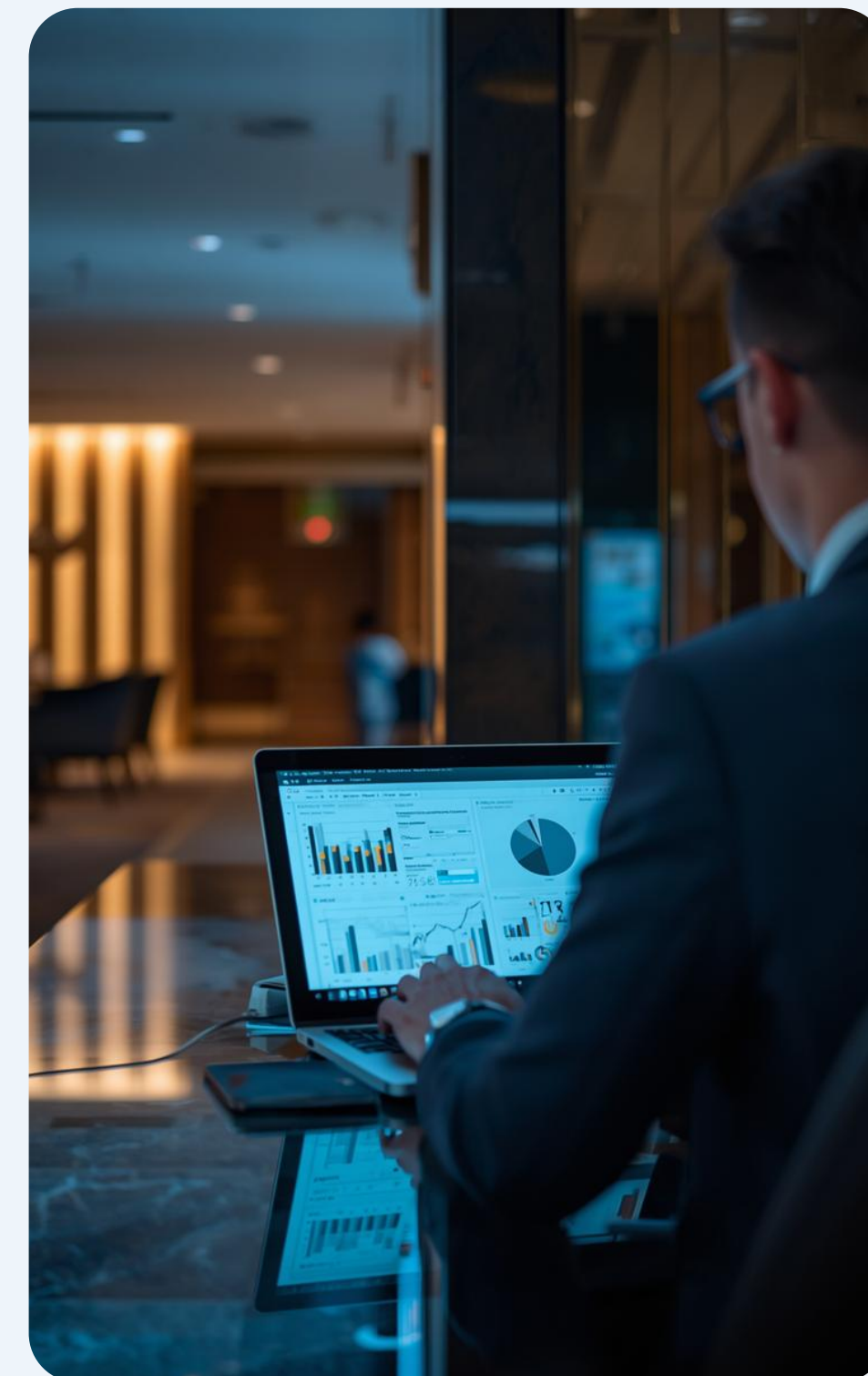
Model accuracy optimization for revenue forecasting using ML hyperparameter tuning

02

Visualization complexity in rendering interactive Plotly & Streamlit dashboard charts

04

Deployment considerations for scalable, real-time Streamlit cloud hosting & performance





Future Scope

Real-Time Data Integration

Connect live booking systems and POS data for instant revenue insights and dynamic forecasting.

Mobile Dashboard

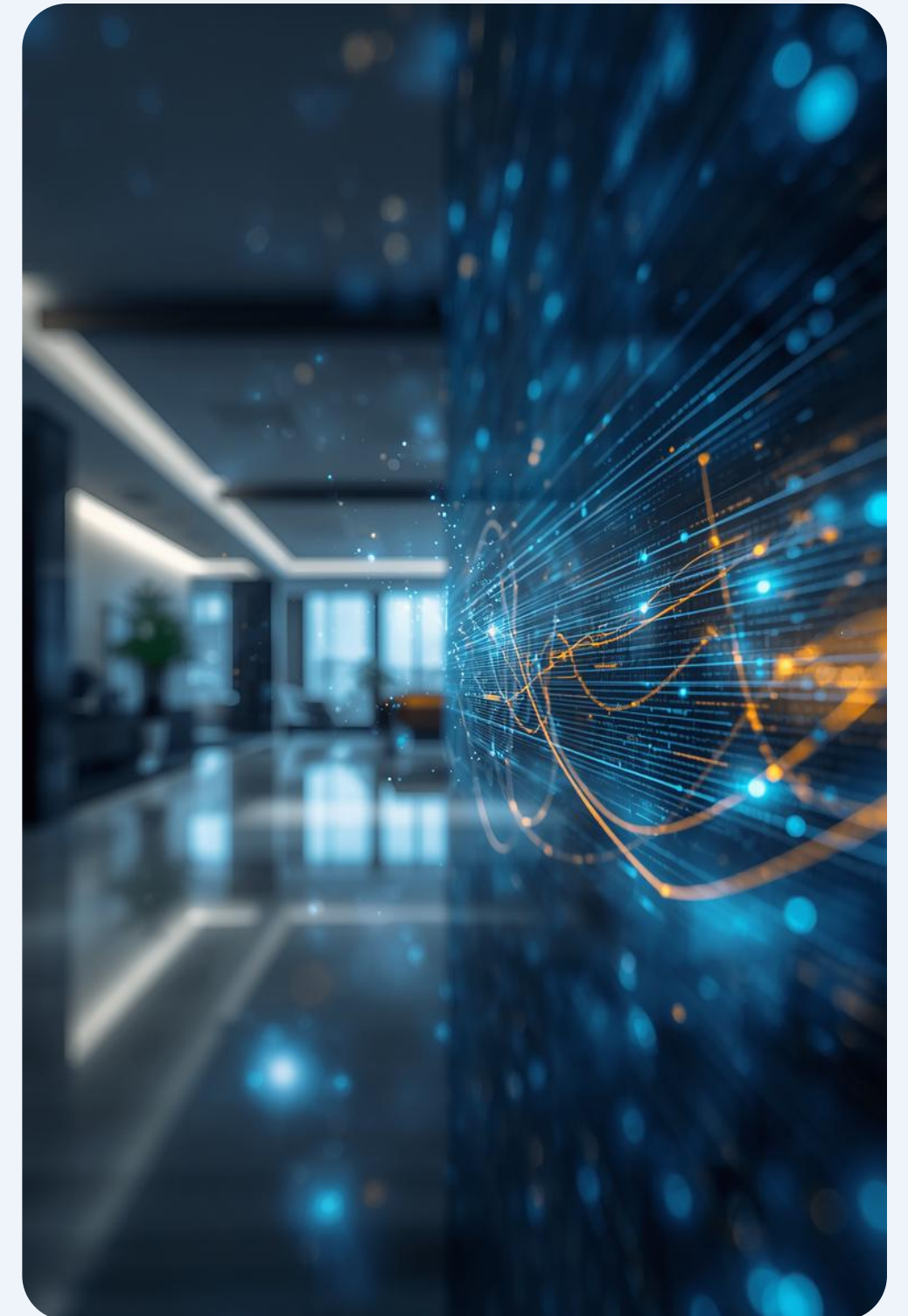
Develop a responsive mobile app for on-the-go access to KPIs, alerts, and revenue reports.

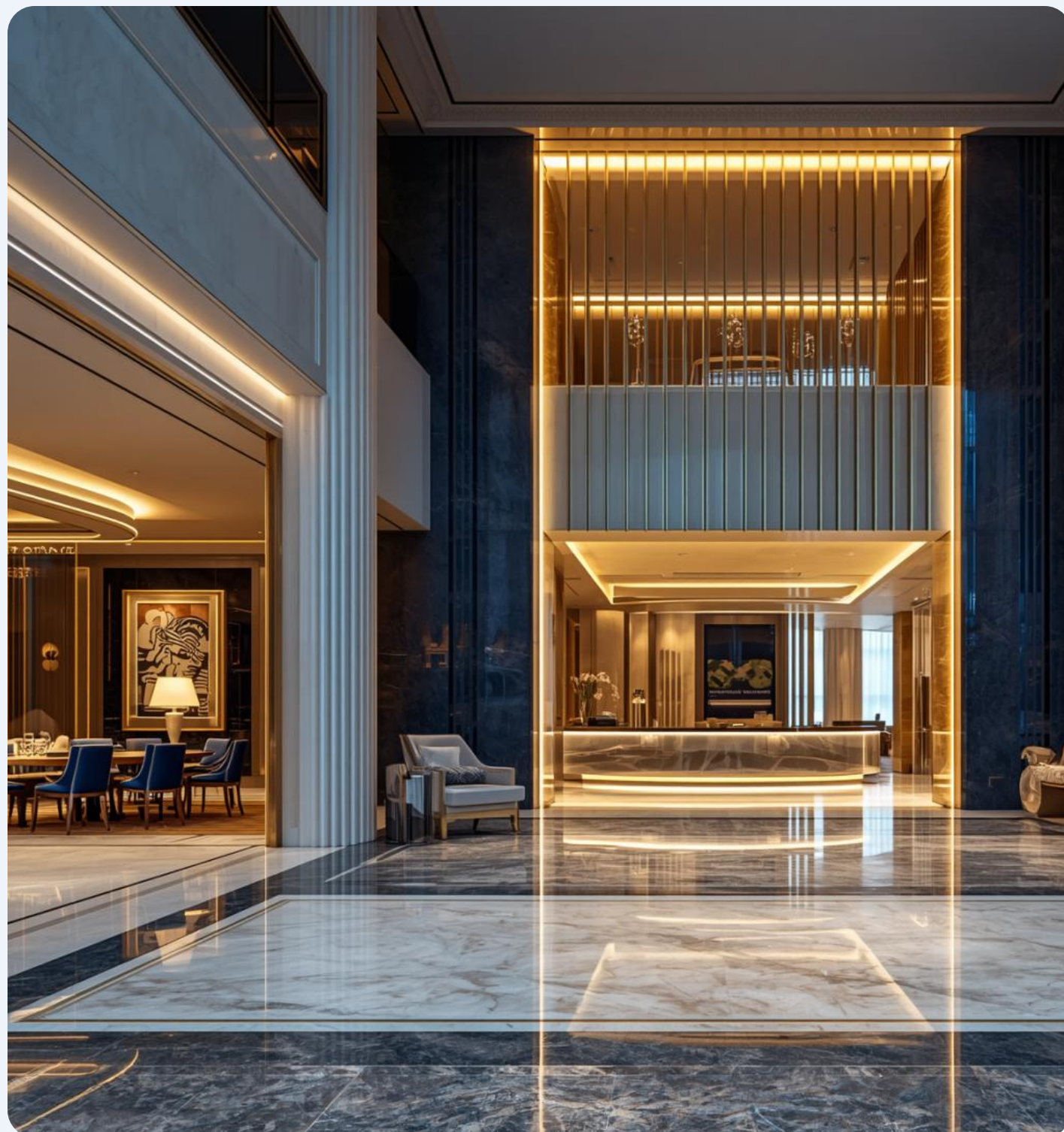
Advanced ML Models

Deploy deep learning and ensemble models for more accurate demand prediction and pricing optimization.

Multi-Property Analysis

Expand the platform to support hotel chains with cross-property benchmarking and consolidated analytics.





Conclusion & Key Takeaways

This project successfully delivered an end-to-end hotel revenue analytics dashboard, transforming raw 2024 data into actionable insights. Using Python, Pandas, Plotly, Streamlit & Machine Learning, we achieved accurate revenue forecasting, occupancy trend analysis, and dynamic pricing recommendations. The solution empowers hotel management to make data-driven decisions, reduce revenue leakage, and optimize operational efficiency — setting a new standard for analytics-driven hospitality management.



Thank You



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